



Analisi e Progetto di Algoritmi

Laurea Triennale in Informatica
(codice: E3101Q113)

AA 2025/2026

**Esistenza di cammini in cui
si alterna il colore degli archi
(variante di Floyd-Warshall)**

Docente: Raffaella Rizzi

[Problema]

INPUT:

- $G = (V, E)$, grafo orientato ~~con archi pesati (pesi positivi)~~,
dove $V = \{1, 2, \dots, n\}$

~~— $W = [w_{ij}]_{n \times n}$, matrice dei pesi~~

- $Col : E \rightarrow C$, con $C = \{\text{"red"}, \text{"green"}\}$

OUTPUT:

per ogni coppia di vertici i e j , verificare se esiste un cammino
in cui si alternano archi rossi e archi verdi

(**NB:** se $i = j$, il cammino $\langle i \rangle$ banalmente verifica
tale condizione)

Sottoproblemi ausiliari per i vertici i e j

Sottoproblema ausiliario di dimensione (k, f, l)

Verificare se esiste un cammino $P_{ij}^{k,f,l}$ da i a j in cui si alternano archi rossi e archi verdi, tale che:

- se $k > 0$ eventuali vertici intermedi appartengono a $\{1, 2, \dots, k\}$ e se $k = 0$ non ci sono vertici intermedi
- il colore del primo arco è f
- il colore dell'ultimo arco è l

$$0 \leq k \leq n \quad f, l \in C$$

Numero di sottoproblemi per i e j : $|C^2|(n+1) = 4(n+1)$

Numero totale di sottoproblemi: $|C^2|(n+1)n^2 = 4(n+1)n^2$



[Coefficienti per i vertici i e j]

Coefficiente del sottoproblema di dimensione (k, f, l)

$e_{ij}^{k,f,l}$ è **TRUE** se e solo se esiste un cammino $P_{ij}^{k,f,l}$ da i a j in cui si alternano archi rossi e archi verdi, tale che:

- se $k > 0$ eventuali vertici intermedi appartengono a $\{1, 2, \dots, k\}$ e se $k = 0$ non ci sono vertici intermedi
- il colore del primo arco è f
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$$0 \leq k \leq n \quad f, l \in C$$

Numero di coefficienti per i e j: $|C^2|(n+1) = 4(n+1)$

Numero totale di coefficienti: $|C^2|(n+1)n^2 = 4(n+1)n^2$

[Eq. Ricorrenza - Casi base]

$k = 0$

$$i = j \quad \Rightarrow e_{ij}^{k,f,l} = \text{FALSE}$$

$$i \neq j \wedge (i,j) \in E$$

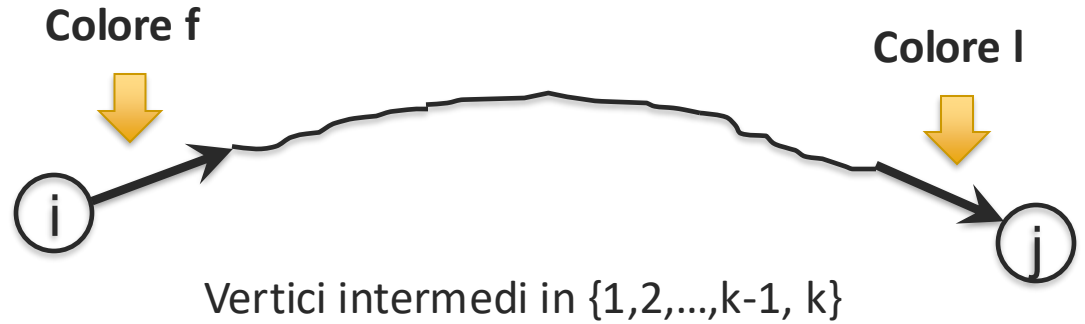
$$f = l \wedge f = \text{Col}(i,j) \quad \Rightarrow e_{ij}^{k,f,l} = \text{TRUE}$$

$$f \neq l \vee f \neq \text{Col}(i,j) \quad \Rightarrow e_{ij}^{k,f,l} = \text{FALSE}$$

$$i \neq j \wedge (i,j) \notin E \quad \Rightarrow e_{ij}^{k,f,l} = \text{FALSE}$$

Eq. Ricorrenza – Passo ricorsivo

$k > 0$



$$e_{ij}^{k,f,l} = ?$$

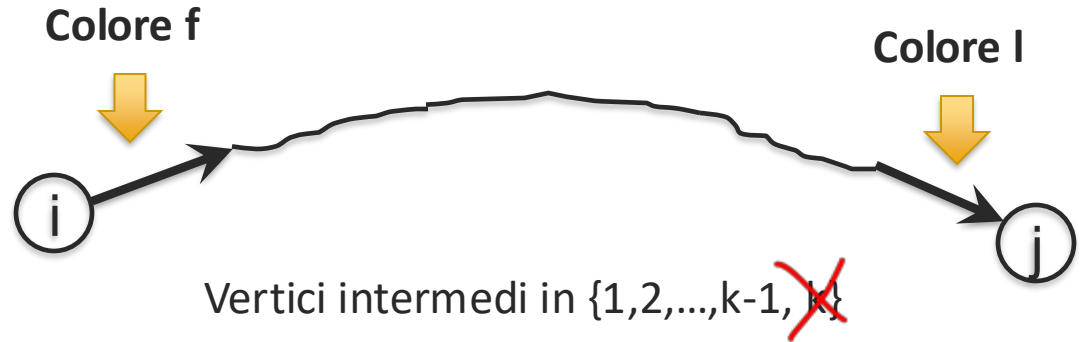
$$\Rightarrow \exists P_{ij}^{k,f,l}$$

Eq. Ricorrenza – Passo ricorsivo

$k > 0$

$k \notin P_{ij}^{k,f,l}$

$\Rightarrow e_{ij}^{k,f,l} = ?$



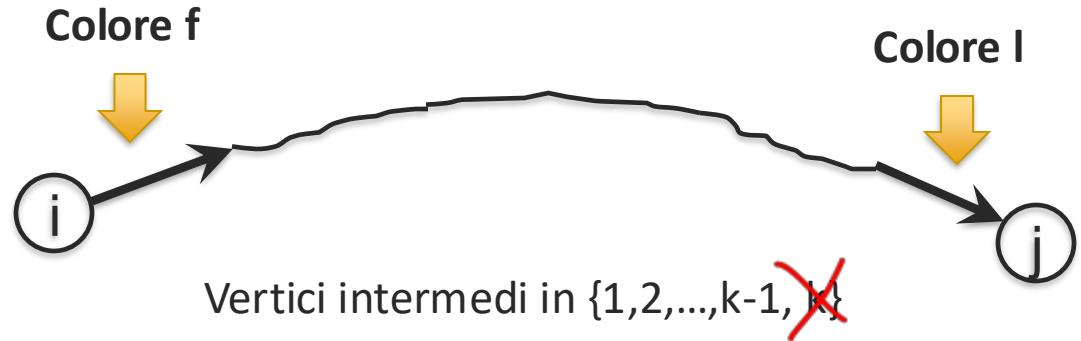
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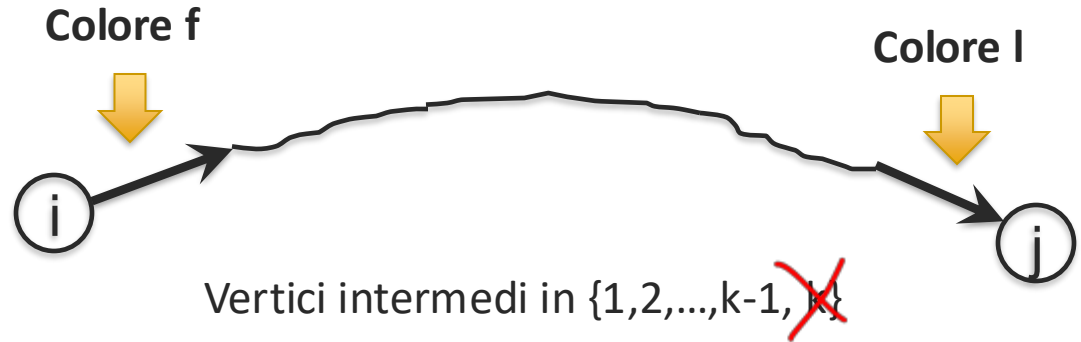
$$\exists P_{ij}^{k-1,f,l} \Rightarrow \exists P_{ij}^{k,f,l}$$

Eq. Ricorrenza – Passo ricorsivo

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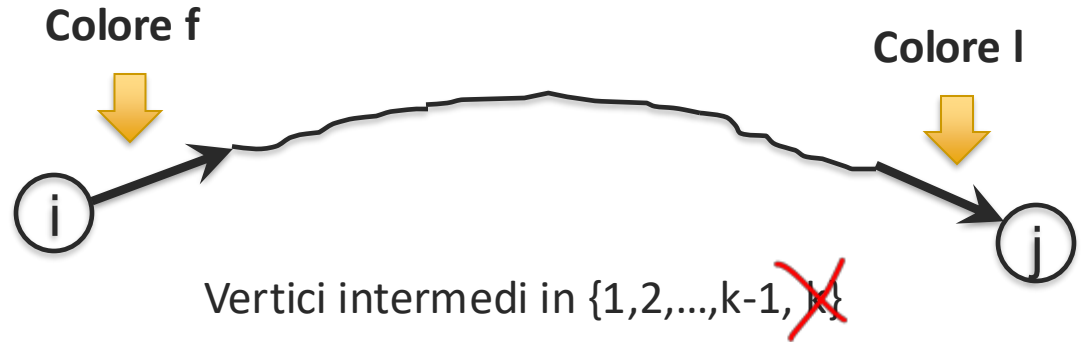
$e_{ij}^{k-1,f,l} \Rightarrow \exists P_{ij}^{k,f,l}$

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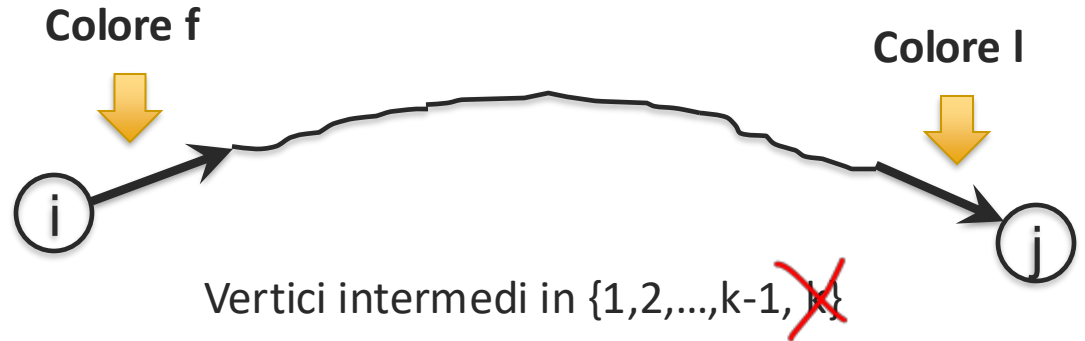
$$e_{ij}^{k-1,f,l} \Rightarrow e_{ij}^{k,f,l}$$

Eq. Ricorrenza – Passo ricorsivo

$k > 0$

$k \notin P_{ij}^{k,f,l}$

$$\Rightarrow e_{ij}^{k,f,l} = e_{ij}^{k-1,f,l} \quad (E1)$$



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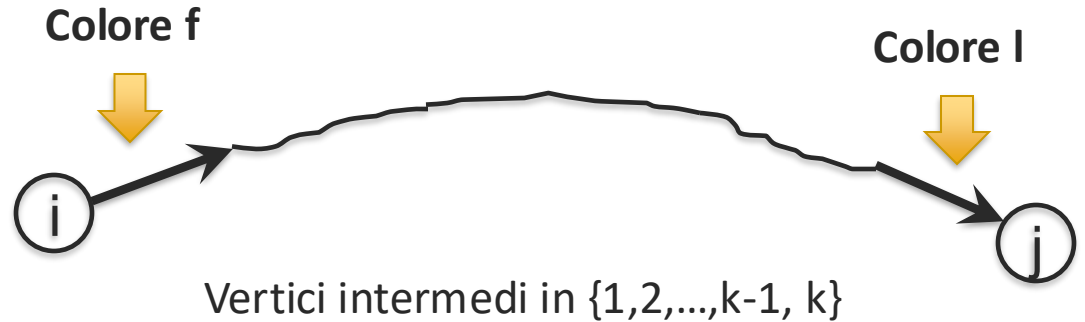
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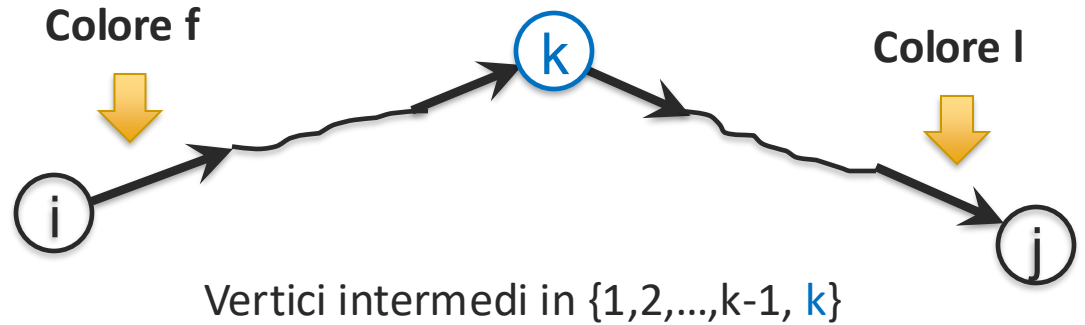
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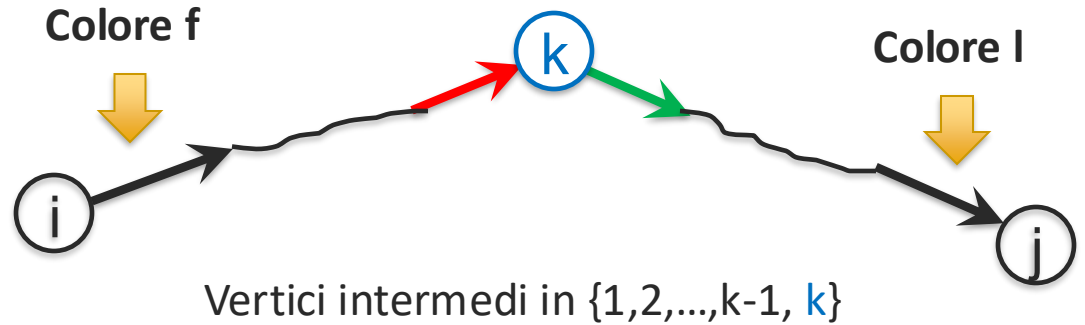
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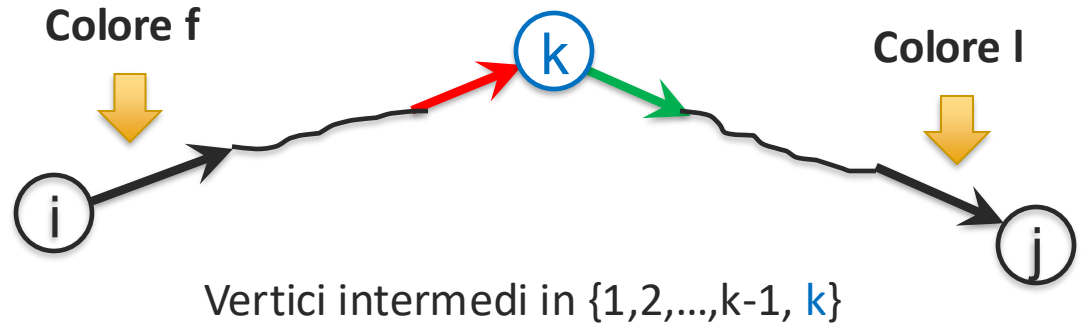
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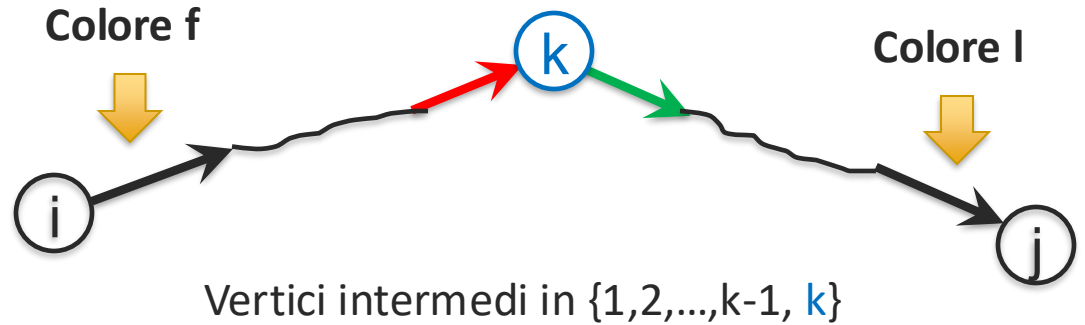
$$\exists P_{ik}^{k-1,f,red} \wedge \exists P_{kj}^{k-1,green,l} \Rightarrow \exists P_{ij}^{k,f,l}$$

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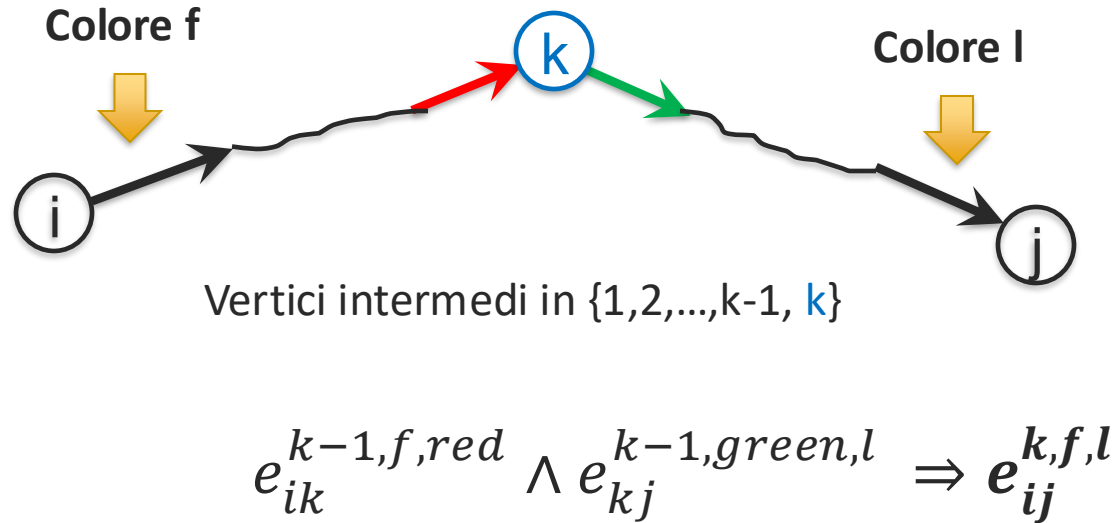
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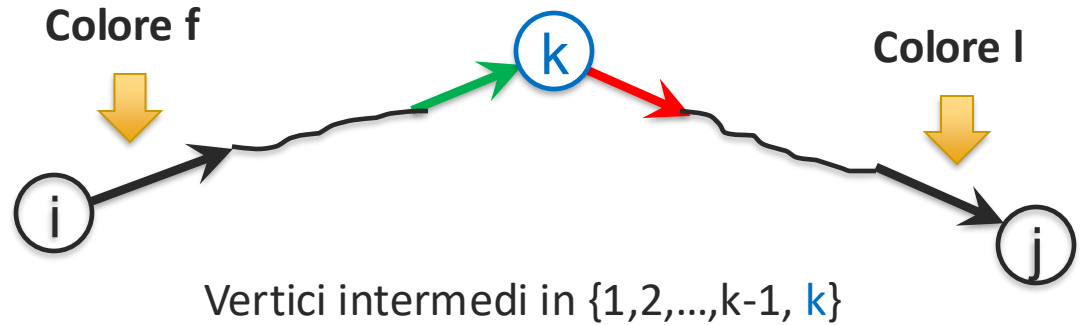
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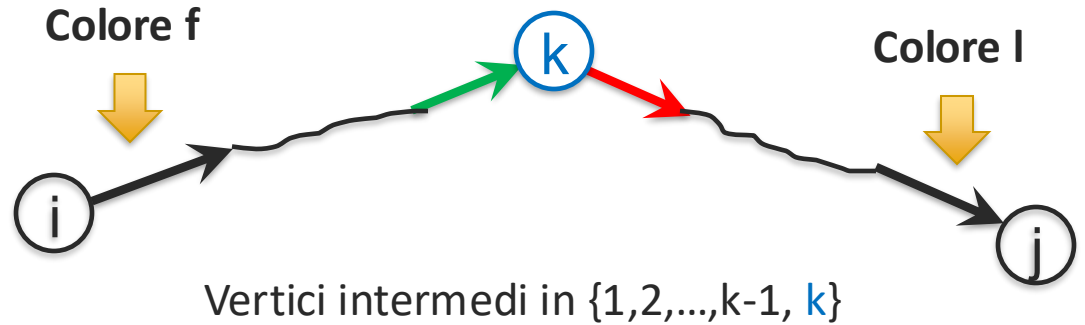
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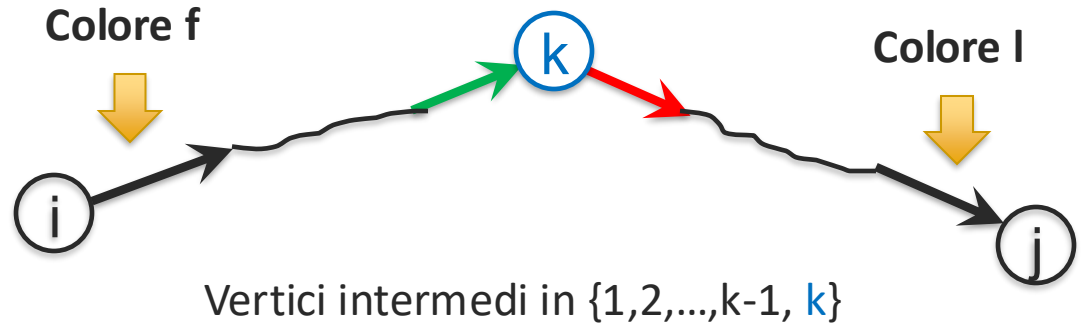
$$\exists P_{ik}^{k-1,f,green} \wedge \exists P_{kj}^{k-1,red,l} \Rightarrow \exists P_{ij}^{k,f,l}$$

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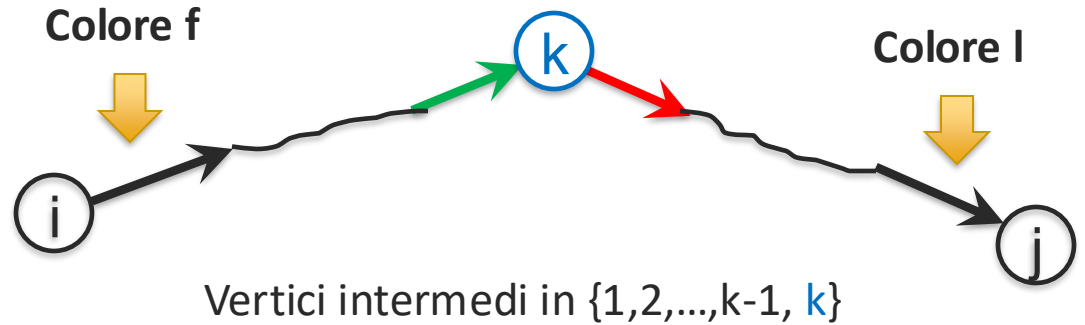
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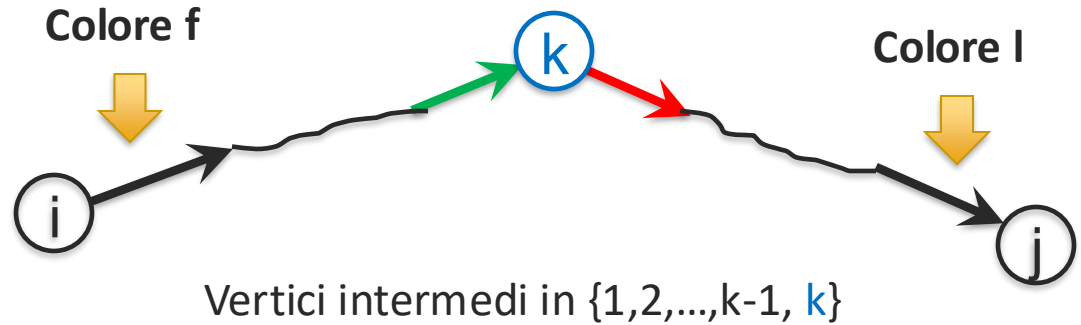
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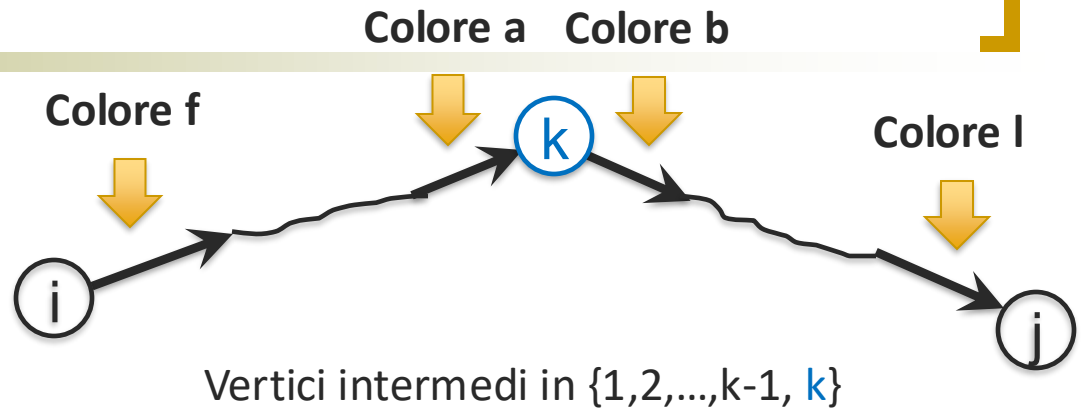
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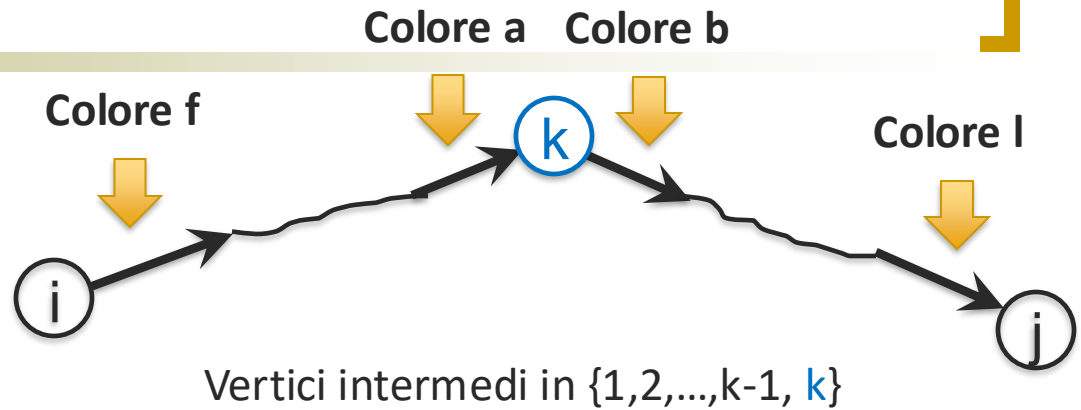
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$$\Rightarrow e_{ij}^{k,f,l} = \vee \{ e_{ik}^{k-1,f,a} \wedge e_{kj}^{k-1,b,l} \mid (a,b) \in C^2 \wedge a \neq b \}$$



Eq. Ricorrenza – Passo ricorsivo

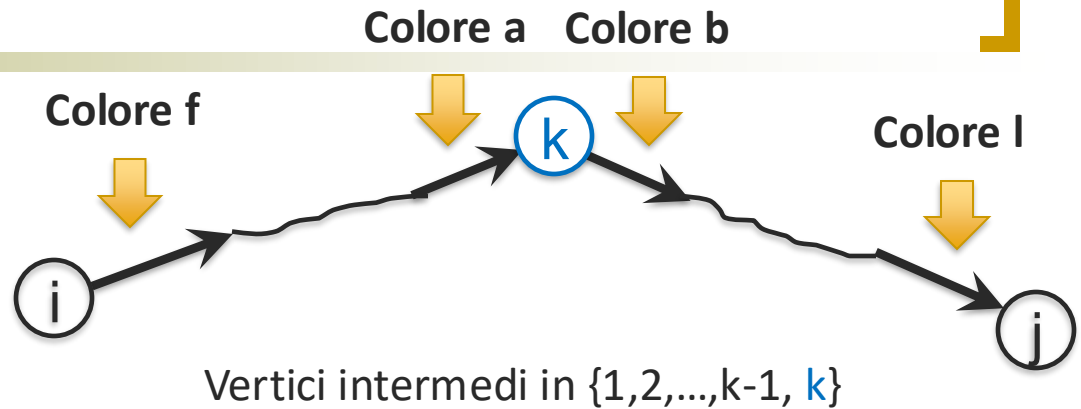
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Eq. Ricorrenza – Passo ricorsivo

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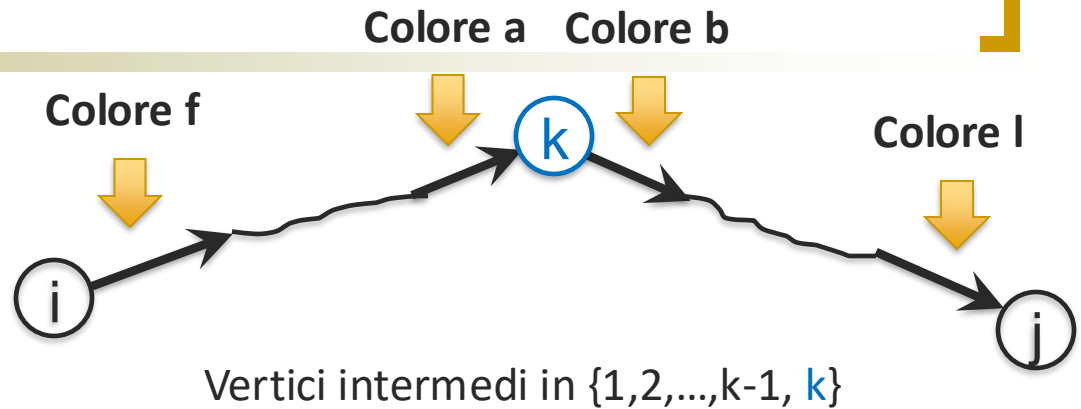
$k \notin P_{ij}^{k,f,l}$

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$$\Rightarrow e_{ij}^{k,f,l} = \vee \{ e_{ik}^{k-1,f,a} \wedge e_{kj}^{k-1,b,l} \mid (a,b) \in C^2 \wedge a \neq b \} \quad (E2)$$

$$e_{ij}^{k,f,l} = E1 \vee E2$$



[Risultato]

$e_{ij}^{n,f,l} \equiv \text{TRUE}$ se e solo se esiste un cammino da i a j in cui si alternano archi rossi e archi verdi, il colore del primo arco è f e il colore dell'ultimo arco è l e i vertici intermedi appartengono a $\{1, 2, \dots, n\}$

[Risultato]

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$$\text{Risultato} = \vee \left\{ \begin{array}{l} e_{ij}^{n,\text{red},\text{red}} \\ e_{ij}^{n,\text{red},\text{green}} \\ e_{ij}^{n,\text{green},\text{red}} \\ e_{ij}^{n,\text{green},\text{green}} \end{array} \right.$$

$$\text{Risultato} = \vee \{ e_{ij}^{n,f,l} \mid (f, l) \in \mathcal{C}^2 \}$$

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$$i \neq j \Rightarrow \text{Risultato} = \vee \{ e_{ij}^{n,f,l} \mid (f, l) \in \mathcal{C}^2 \}$$

[Risultato]

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$i = j \Rightarrow \text{risultato} = \text{TRUE}$